Muhammad Andrata Zharfan Mustika

AI Engineer | Data Scientist | Data Analyst
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SUMMARY

Recent Informatics Engineering graduate targeting AI Engineer, Data Scientist, or Data Analyst roles, with experience building LLM/RAG systems and machine learning models in Python. The thesis RAG question-answering system raised Answer Correctness from 0.53 to 0.73 versus a no-RAG baseline (80 questions validated by a nutrition expert). Accustomed to translating business needs into technical solutions as the point of contact for stakeholders and clients.

EDUCATION

Universitas Komputer Indonesia (UNIKOM), Bandung *2022 - 2026*
Bachelor of Computer Science (S.Kom.), Informatics Engineering
GPA: 3.75 / 4.00

WORK EXPERIENCE

Fullstack Web Developer Intern *Aug 2025 - Nov 2025*
Rama Group (Rama FM & Paramuda Radio)

- Led a team of 6 interns in modernizing the UI/UX of two radio websites (Rama FM and Paramuda Radio) using Laravel, Inertia, and React, through to release within the 4-month internship (development completed in 3 months).
- Built 10+ features (streaming, replays, Paramuda live video, programs and hosts, articles, music chart, magazine/events) with the interface tailored to each target market: adult and senior listeners for Rama FM, young people for Paramuda.
- Provided visitor data that previously did not exist through an analytics dashboard (daily visits, hourly activity, top 10 cities and devices, top 5 most-read articles) and a WhatsApp contact widget managed from the admin panel, making the marketing team's promotional agenda data-driven.
- Translated business needs into website features as the technical point of contact for stakeholders and across 5+ client meetings with the marketing team, including proposing a Rama FM birthday greeting card that the radio station approved, so employees, clients, and listeners can send greetings through the website.

ORGANIZATIONAL EXPERIENCE

General Secretary *2024 - 2025*
Informatics Engineering Student Association (HMIF) UNIKOM

- Oversaw administration for 94 members, 7 divisions, and 8 mega events by drafting 88 organization-wide letters and screening 108 event letters from event secretaries, so that all letters were issued in line with administrative standards and a uniform format.
- Improved the Organizational Administration Implementation Guidelines to be neater and clearer, with added usage examples for letters, proposals, and Accountability Reports (LPJ), so event secretaries understood administrative procedures about 2x faster without repeated verbal explanations.
- Performed the role independently while keeping the chairman continually updated, ensuring directives reached all 7 divisions and every member, and supporting all activities so that each division's progress was monitored and obstacles were reported quickly.

PROJECTS

Implementation of Retrieval-Augmented Generation (RAG) With Qwen 2.5 7B Instruct Generator in a Stunting Question-Answering System (Thesis) *2025 - 2026*
Python · LLM · LangChain · RAG · RAGAS · FAISS · Gradio

- Built an end-to-end Indonesian-language RAG pipeline to answer public health questions from 60 official documents & journals, indexing 12,558 structured chunks using multilingual-e5-base and the FAISS vector database.
- Developed a Gold Standard evaluation dataset (80 QA pairs) across 6 question categories, validated directly with a nutrition expert through a Human-in-the-Loop (HITL) approach.

- Raised Answer Relevancy from 0.73 to 0.92 and Answer Correctness from 0.53 to 0.73 compared with a no-RAG baseline (same 80 questions), with Retrieval MRR 0.95, Context Precision 0.96, and Faithfulness 0.90 (RAGAS); the main weakness is numerical questions (Faithfulness 0.69).

Early Detection of Heart Disease Patients *2026*

Python · XGBoost · SHAP · MLJAR AutoML

- Produced an early heart disease detection model with 84.1% accuracy and 87.5% recall on the test set (Decision Tree, max_depth=3, best of 4 models), on 908 patient records (11 features), through SHAP-based feature selection and comparison against Random Forest, Logistic Regression, and AutoML.

Classification of Child Stunting Status from Basic Anthropometric Data *2026*

Python · Scikit-learn · Pandas · Seaborn

- Compared Random Forest and Logistic Regression on 92,692 synthetic records, and identified that the near-perfect score (macro-F1 ≈ 1.00) comes from labels derived from z-score rules, not real generalization.

Sentiment Analysis of the Film Jumbo on Twitter (X) *2025*

Python · Scikit-learn · Pandas · Sastrawi · NLTK · imbalanced-learn

- Built a three-class sentiment classification pipeline on 5,302 Indonesian-language tweets (CRISP-DM), covering preprocessing (slang normalization, Sastrawi stemming, negation handling), TF-IDF unigram-bigram, and Logistic Regression.
- Raised negative-class recall (only 5.4% of the data) from 0.12 to 0.58, with 69% accuracy and macro-F1 0.63 on the test set, using class weights, SMOTE, RandomOverSampler, and 5-fold GridSearchCV.

Face Image Classification with CNN (ResNet50) *2025*

Python · PyTorch · Convolutional Neural Networks · OpenCV · Scikit-learn

- Built a three-class face image classification pipeline on 86,744 images, covering blurry-image filtering with Laplacian variance (32,648 images remaining), face detection and cropping with Haar Cascade, data augmentation, and class imbalance handling with WeightedRandomSampler.
- Achieved 85.76% validation accuracy and macro-F1 0.85 (about 18 points higher than training the final layer only) through two-stage fine-tuning of an ImageNet-pretrained ResNet50.

SKILLS

AI and Data: Large Language Model (LLM), LLM API, Retrieval-Augmented Generation (RAG), Vector Database (FAISS), Machine Learning, Feature Engineering, Natural Language Processing (NLP), Deep Learning, Computer Vision, Predictive Modeling, Prompt Engineering, Model Evaluation (RAGAS), Feature Selection (SHAP), Sentiment Analysis, Imbalanced Data Handling, Data Preprocessing, Exploratory Data Analysis (EDA), Data Analysis, Data Visualization. Programming and Databases: Python, SQL, MySQL, JavaScript, PHP.

Programming and Databases: Python, SQL, MySQL, JavaScript, PHP.

Frameworks and Tools: Pandas, Scikit-learn, PyTorch, LangChain, Hugging Face Transformers, XGBoost, OpenCV, NLTK, Seaborn, Gradio, Jupyter Notebook, Git, Laravel, React, Google Workspace, Microsoft Office.

Soft Skills: Analytical and Critical Thinking, Problem Solving, Data Storytelling, Understanding of Business Needs, Data-Driven Decision Making, Communication, Presentation, Public Speaking (MC, moderator, speaker), Teamwork, Leadership, Time Management and Prioritization, Adaptability, Documentation.

Languages: Indonesian; English (TOEFL ITP: 550).

CERTIFICATIONS & COURSES

Associate Data Scientist (CADS) - BNSP *2026*
Gemini Certified University Student - Google *2026*
Artificial Intelligence & Automation Bootcamp - MySkill *2026*
AI for Work & Career Readiness With Google AI Products - Hacktiv8 *2026*
Data Analyst Fundamental - MySkill *2026*
Google AI Essentials - Google *2025*
Software Engineer Fundamental Course - RevoU *2024*